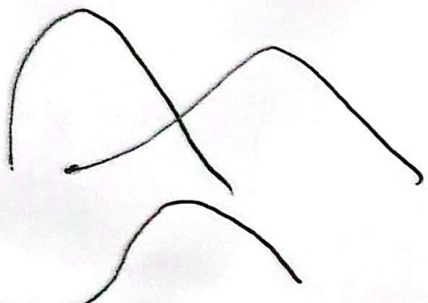


Term test 2
Principles of Microeconomics
ECO0311 21010(STA)
Time: 30 min, Marks: 20

- ✓ 1. Explain **consumer surplus** and **producer surplus** using a demand and supply diagram. (8)
- ✓ 2. Discuss the impact of a **price ceiling** on market equilibrium. (6)
- ✓ 3. Suppose the demand function for a product is $Q_d = 100 - 5P$ and the supply function is $Q_s = 20 + 3P$. Find the **equilibrium price and quantity**. (6)



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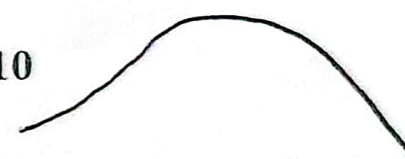
2nd year 1st Semester Examination 2024

Course No.: STA 0542 2153

2nd Term Test

Time: 30 Mins.

Full Marks: 10



Answer all of the following questions

1. Define Student's t-statistic. Find the moments of t-distribution and hence find mean, variance, skewness and kurtosis. 6
2. Establish the relationship between t and F distribution. 4

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2nd year 1st Semester Examination 2024

Course No.: STA 0542 2153

1st Term Test

Full Marks: 20

$$f(x_1, x_2) =$$

Answer all of the following questions

- ✓ 1. Let X be a continuous random variable with density function

$$f(x) = \frac{x}{2}, 0 < x < 2$$

3

Find the distribution of $Z=2X+3$

- ✓ 2. Let X_1 and X_2 be two independent standard normal random variables. Let $Y_1=X_1+X_2$ and $Y_2=X_2-X_1$. Find the joint distribution of Y_1 and Y_2 .

4

3. If $X \sim \chi^2_{(n)}$, show that mode is at $x = n-2$.

- ✓ 4. Let X and Y be two independent random variables having chi-square distribution with degrees of freedom m and n respectively. Obtain the distribution of

$$U = \frac{X}{X+Y}$$

$$\sqrt{\frac{m+n}{2}}$$

6/22

Test 2, MAT 05412105

Answer any two, Time 40 min

1. If $\mathcal{L}\{F(t)\} = f(s)$, then show that $\mathcal{L}\{t^n F(t)\} = (-1)^n f^{(n)}(s)$, $n = 1, 2, 3, \dots$ and hence evaluate $\int_0^\infty t e^{-3t} \cos t \, dt$.
2. Show that $\int_0^\infty \frac{\sin t}{t} \, dt = \frac{\pi}{2}$.
3. Use Laplace transformation to solve the IVP: $Y'' - 3Y' + 2Y = 4e^{3t}$, $Y(0) = 3$, $Y'(0) = 4$.

$$\int_s^\infty f(u) \, du$$

MAT05412105; Advanced Calculus and Differential Equations

Answer any two; Time: 35 min

- ✓ 1. When is a 1st order and 1st degree differential equation said to be homogeneous? State whether the following equation is homogeneous or not and then solve it: $(x^2 - 3y^2)dx + 2xydy = 0$.
2. Define an exact differential equation. Write down the necessary and sufficient condition that a 1st order and 1st degree differential equation to be exact. Solve the initial value problem: $(2xy - 3)dx + (x^2 + 4y)dy = 0$, $y(1) = 2$.
- ✓ 3. Write down the general form of a 1st order linear equation. Show that a linear equation is not exact. Find the integrating factor to make the following linear equation into an exact equation and then solve: $\frac{dy}{dx} + \left(\frac{2x+1}{x}\right)y = e^{-2x}$.

$$\frac{dy}{dx} + \frac{1}{x}$$

MAT05412105; Advanced Calculus & Differential equations

Answer any two; Time: 40 min

$$\frac{1}{\sqrt{n^2 - a^2}}$$

8/77

1. Define improper integral of type I and type II. Evaluate $\int_4^{16} \frac{1}{\sqrt{x^2-16}} dx$.
2. Show that $\int_0^{\pi/2} \sin^m x \cos^n x dx = \frac{\Gamma(\frac{m+1}{2}) \Gamma(\frac{n+1}{2})}{2 \Gamma(\frac{m+n+2}{2})}$ & hence evaluate $\int_0^{\pi/2} \sin^3 x \cos^{5/2} x dx$.
3. State the method of Lagrange Multipliers to find the local extreme values of a function subject to a constraint. Find the maximum and minimum values of the function $f(x, y) = 3x + 4y$ on the circle $x^2 + y^2 = 1$.

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2nd Year 1st Semester 2024(Term Test-I)

Course No. STA 0541 2151

Full Marks: 20, Time: 40 Minutes

- ✓ 1. Define (i) probability function in terms of algebra of events and (ii) probability space. 4
- ✓ 2. Distinguish between mass function and density function of random variables? Let $f(x) = K(2x - x^2); 0 < x < 2$. 6
3/4
(i) Find K such that $f(\cdot)$ is a density function.
(ii) Find the corresponding c.d.f.
(iii) Find $P[X \leq 1]$.
$$\int_0^2 K(2x - x^2) dx = 1$$
$$\Rightarrow K \left[\frac{2x^2}{2} - \frac{x^3}{3} \right]_0^2 = 1$$
- ✓ 3. A machine has a probability of 0.85 of functioning correctly for each hour. ~~Given that the machine has already operated successfully for 20 hours,~~ what is the probability that it malfunctions for the first time exactly after 26 hours of operation? Also, find the mean and variance of the total number of hours the machine functions before its first malfunction. 4
(0.15)(0.85)²⁵
- ✓ 4. Compare and contrast the binomial distribution and the Negative Binomial Distribution. 6

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2nd Year 1st Semester 2024(Term Test-II)

Course No. STA 0541 2151

Full Marks: 20, Time: 45 Minutes

[Answer any three questions]

-
1. Compare and contrast the hypergeometric and binomial distributions, highlighting their key differences and applications.
 2. Let X denotes the number of failures to get r successes when sampling without replacement from a population of size N containing M successes. Then X [$X=0, 1, \dots, N-M$] has a negative (or inverse) hypergeometric distribution (NHGD) with parameters M, r, N denoted as $NHG(M, r, N)$. Derive the probability mass function (PMF) of X .
 3. Explain Beta-Binomial Distribution as an Urn Model and link it with the broader family of probability distributions.
 4. Derive the PMF of a generalized negative binomial distribution.